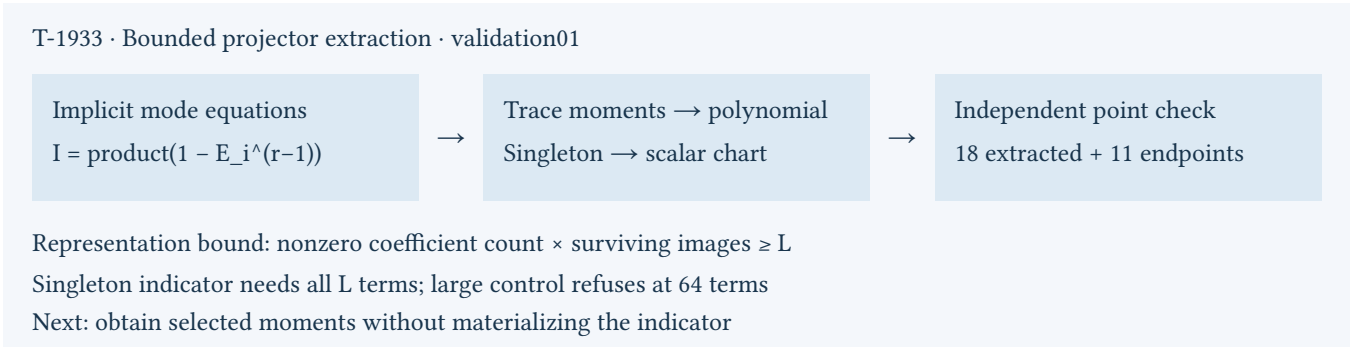
Bounded projector extraction

T-1933 · Prime fields · 2026-11-08

The optional extractor obtains candidate images from the implicit three-mode constraints, without a supplied Y or runtime candidate scan. It constructs a support indicator, reads trace moments and applies Newton identities.



Derived construction. In the quotient by Y^L−epsilon, the product of $1-E_i^{(r-1)}$ is1 at common roots and0 elsewhere. If its coefficients are c_e, the survivor count is Lc_0 and the j-th moment is L epsilon c(L−j), with the stated endpoint convention. Newton identities reconstruct the exact candidate polynomial. A singleton uses the existing rational scalar chart and an independent public-point check.

Recorded qualification. All528 subset projectors and22 dense singleton controls pass. All144 transcript annihilators match the direct oracle, with1,176 phase memberships checked. Maxima for those144 cases are67,512 entered monomial pairs,15,558 retained node-coefficient slots and20 temporary terms. These are separate dimensions, not total work or peak RAM.

Public recovery. On29 fixed public targets,18 recover from extracted singleton images and11 from verified identity/endpoint exits. None is unresolved or capped. No fixture Y is passed to the adapter. Nominal recovery uses332 API queries and689,538 projector pairs;27 passing call guards add27 queries (359 including guards). Source setup/calibration and other operations remain charged. PCC-20261108-80d4d8 is unsigned producer evidence.

Preserved large refusal. Fixture observations at prime998244353, m7 and L71,303,168 hit the64-term cap after14,621 entered pairs and5,724 retained slots. This is not a large public-point recovery. The refusal and four separate cap controls are retained; no dense fallback runs.

Representation bound. Multiplication by the indicator has rank d, the number of survivors. Its cyclic-shift columns each have s nonzero coefficients. A basis of d columns must collectively cover all L coordinates, hence sd≥L. In particular singleton support requires all L coefficients. This bounds the explicit indicator, not every implicit extractor.

Next step and costs. Obtain selected trace moments without materializing the indicator. Low-degree truncation alone is invalid: high exponents wrap into low coefficients. Charge source operations, point shifts, map lookups, copies, allocations, coefficient updates, Newton inversions and verification, in addition to recorded monomial pairs.

Custody and sources. Fresh overflow-checked build/check pass within90/60 seconds; ten copied files and15 dependencies are hash-bound. Primary source: Chen-Fan-Luo, arXiv2211.11204v3, finite-field support/rank discussion. DERIVATION.md gives the direct quotient proof. KB unavailable; no novelty/timing claim or cloud submission. Independent signed review pending.